

# Wave 2 Statistical Analyses

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## Hierarchical Bayesian Models Wave 2

This document runs the models used to analyse the main outcomes for Wave 2.  
The structure of the document follows the Analysis\_Plans.txt file.  
The variables in the datasets are explained in Meta-Data.txt.

### Data Set 1 - Binary Categorical Outcomes

#### Crisis aware

Bayesian Logistic regression for whether participants were aware of the reproducibility crisis or not (yes/no).  
The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement or third year
- \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu              0.39     0.01 0.37   -0.31    0.14    0.39    0.63    1.13  4668
## sigma           0.88     0.00 0.21    0.55    0.74    0.86    1.00    1.36 10403
## alpha[1]        -0.60     0.01 0.70   -1.96   -1.06   -0.59   -0.14    0.79  6167
## alpha[2]         0.20     0.01 0.75   -1.27   -0.29    0.19    0.69    1.72  7257
## alpha[3]         1.61     0.01 0.72    0.26    1.13    1.59    2.06    3.07  6122
## alpha[4]         0.08     0.00 0.27   -0.45   -0.11    0.08    0.26    0.61  5378
## alpha[5]         0.98     0.01 0.60   -0.16    0.58    0.96    1.38    2.20 10136
## alpha[6]        -0.39     0.01 0.52   -1.45   -0.73   -0.38   -0.04    0.60  9755
## alpha[7]        -0.32     0.01 0.43   -1.19   -0.61   -0.32   -0.03    0.51  6718
## alpha[8]         1.33     0.01 0.49    0.41    0.99    1.31    1.65    2.33  7248
## alpha[9]         0.64     0.00 0.33   -0.01    0.42    0.64    0.86    1.29  9734
## alpha[10]       -0.22     0.00 0.45   -1.12   -0.51   -0.21    0.09    0.65  8364
## alpha[11]        0.44     0.01 0.56   -0.64    0.06    0.43    0.81    1.57 11509
## alpha[12]        1.49     0.01 0.69    0.26    1.01    1.45    1.92    2.94  9288
## alpha[13]        0.43     0.01 0.51   -0.57    0.09    0.43    0.77    1.42  9275
## alpha[14]        0.04     0.01 0.58   -1.09   -0.35    0.04    0.43    1.15 12346
## alpha[15]        1.38     0.01 0.49    0.46    1.04    1.37    1.70    2.37  7239
## alpha[16]       -0.17     0.01 0.55   -1.28   -0.53   -0.16    0.20    0.90 11316
## alpha[17]       -0.31     0.01 0.53   -1.38   -0.67   -0.30    0.05    0.70  9167
## alpha[18]       -0.13     0.01 0.53   -1.20   -0.48   -0.12    0.24    0.90 10171
## alpha[19]        0.95     0.01 0.53   -0.08    0.59    0.94    1.30    2.01  6812
## beta1          -1.34     0.01 0.85   -3.05   -1.90   -1.34   -0.77    0.31  7083
```

```

## beta2      -0.17    0.00 0.15   -0.46   -0.27   -0.17   -0.07    0.11  4007
## beta3       1.37    0.00 0.33    0.74    1.14    1.36    1.59    2.02  9929
## lp__      -411.54    0.05 3.88 -420.07 -413.94 -411.19 -408.75 -404.92  6747
##           Rhat
## mu          1
## sigma        1
## alpha[1]     1
## alpha[2]     1
## alpha[3]     1
## alpha[4]     1
## alpha[5]     1
## alpha[6]     1
## alpha[7]     1
## alpha[8]     1
## alpha[9]     1
## alpha[10]    1
## alpha[11]    1
## alpha[12]    1
## alpha[13]    1
## alpha[14]    1
## alpha[15]    1
## alpha[16]    1
## alpha[17]    1
## alpha[18]    1
## alpha[19]    1
## beta1        1
## beta2        1
## beta3        1
## lp__         1
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 01:15:11 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

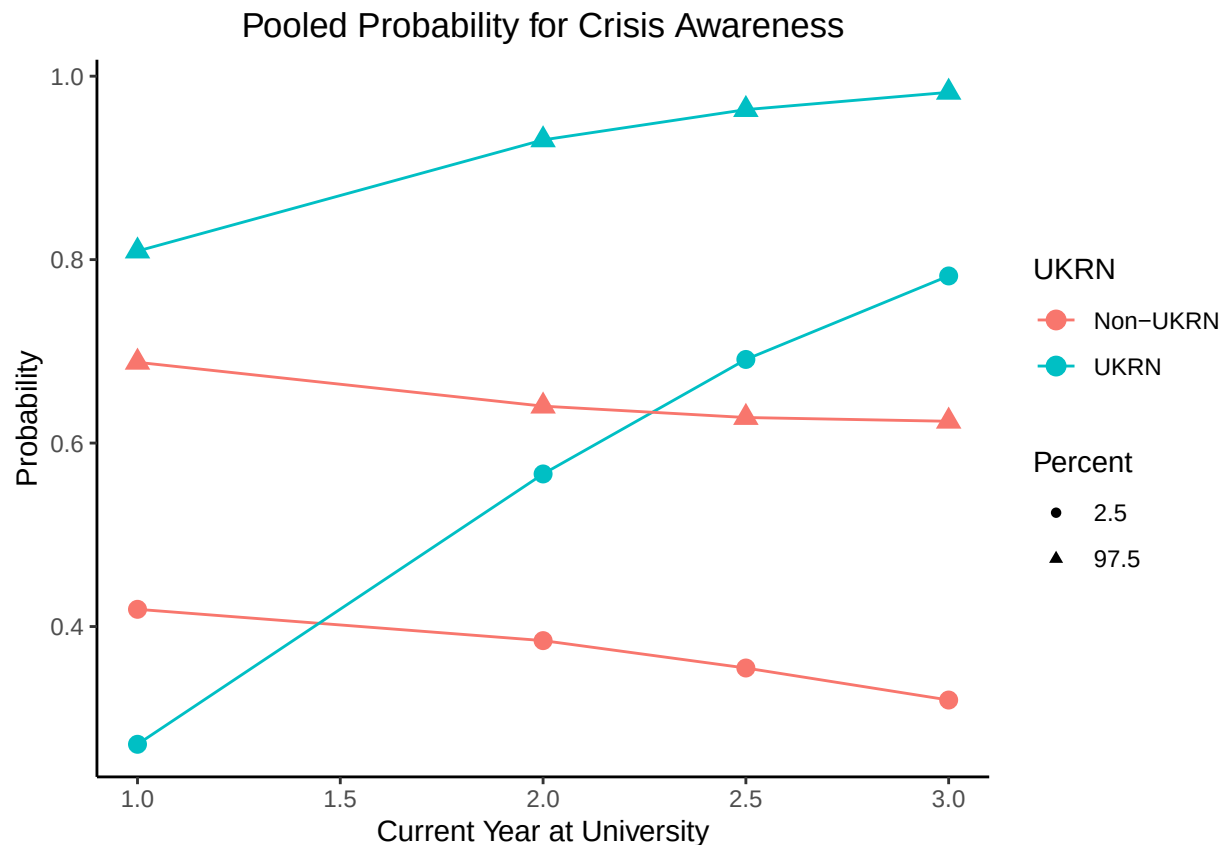
```

#### Number of participants in the analysis

**Total N** = 682  
**UKRN N** = 221  
**Non-UKRN N** = 461  
**Year 1 N** = 308  
**Year 2 N** = 255  
**Placement Year N** = 31  
**Year 4 N** = 88

The output shows that beta credibility intervals do not cross 0 only for the interaction term.  
 2.5% = 0.7382417  
 97.5% = 2.0233802

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of the reproducibility crisis in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.



### Learned about Open Science at Uni

Bayesian Logistic regression for whether participants received teaching about open science or not (yes/no).  
The explainer variables are:

\* Whether the participant was at a university that was affiliated with the UKRN

\* Whether the participant was in their first, second, placement of third year \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean  sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           -1.56    0.01  0.40   -2.36   -1.82   -1.55   -1.29   -0.80  3462
## sigma         0.79    0.00  0.21    0.45    0.64    0.76    0.90    1.28  7959
## alpha[1]     -2.30    0.01  0.65   -3.64   -2.72   -2.28   -1.86   -1.05  4705
## alpha[2]     -1.39    0.01  0.69   -2.78   -1.83   -1.38   -0.93   -0.05  5482
## alpha[3]     -1.01    0.01  0.65   -2.31   -1.42   -1.00   -0.59    0.27  4503
## alpha[4]     -0.50    0.00  0.31   -1.09   -0.71   -0.50   -0.29    0.11  4457
## alpha[5]     -1.21    0.01  0.63   -2.48   -1.63   -1.21   -0.78    0.02  8544
## alpha[6]     -2.10    0.01  0.68   -3.56   -2.53   -2.06   -1.64   -0.87  7930
## alpha[7]     -2.07    0.01  0.57   -3.26   -2.44   -2.05   -1.68   -1.02  6273
## alpha[8]     -2.13    0.01  0.61   -3.41   -2.52   -2.11   -1.71   -1.02  7139
## alpha[9]     -2.10    0.01  0.48   -3.10   -2.41   -2.08   -1.77   -1.23  8650
## alpha[10]    -2.06    0.01  0.60   -3.31   -2.44   -2.03   -1.64   -0.97  7628
## alpha[11]    -1.86    0.01  0.69   -3.31   -2.29   -1.83   -1.38   -0.58  8947
```

```

## alpha[12]    -0.79    0.01 0.64   -2.06   -1.22   -0.79   -0.36    0.49  9150
## alpha[13]    -1.22    0.01 0.58   -2.38   -1.60   -1.22   -0.83   -0.11  7527
## alpha[14]    -1.81    0.01 0.69   -3.26   -2.24   -1.78   -1.34   -0.54  9061
## alpha[15]    -1.08    0.01 0.51   -2.09   -1.42   -1.08   -0.74   -0.08  5926
## alpha[16]    -1.90    0.01 0.69   -3.36   -2.32   -1.86   -1.43   -0.65  8162
## alpha[17]    -1.50    0.01 0.60   -2.71   -1.90   -1.49   -1.11   -0.35  7301
## alpha[18]    -1.97    0.01 0.67   -3.42   -2.38   -1.94   -1.51   -0.75  7994
## alpha[19]    -0.67    0.01 0.56   -1.78   -1.05   -0.66   -0.29    0.43  5943
## beta1        -0.10    0.01 0.80   -1.72   -0.63   -0.10    0.43    1.46  5323
## beta2        -0.09    0.00 0.17   -0.43   -0.20   -0.09    0.03    0.25  3555
## beta3         0.74    0.00 0.31    0.14    0.53    0.73    0.94    1.34  7069
## lp__         -362.16   0.06 4.30 -371.47 -364.83 -361.84 -359.11 -354.59 5033
##              Rhat
## mu              1
## sigma           1
## alpha[1]        1
## alpha[2]        1
## alpha[3]        1
## alpha[4]        1
## alpha[5]        1
## alpha[6]        1
## alpha[7]        1
## alpha[8]        1
## alpha[9]        1
## alpha[10]       1
## alpha[11]       1
## alpha[12]       1
## alpha[13]       1
## alpha[14]       1
## alpha[15]       1
## alpha[16]       1
## alpha[17]       1
## alpha[18]       1
## alpha[19]       1
## beta1           1
## beta2           1
## beta3           1
## lp__            1
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 01:18:25 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

#### Number of participants in the analysis

Total N = 679

UKRN N = 220

Non-UKRN N = 459

Year 1 N = 308

Year 2 N = 253

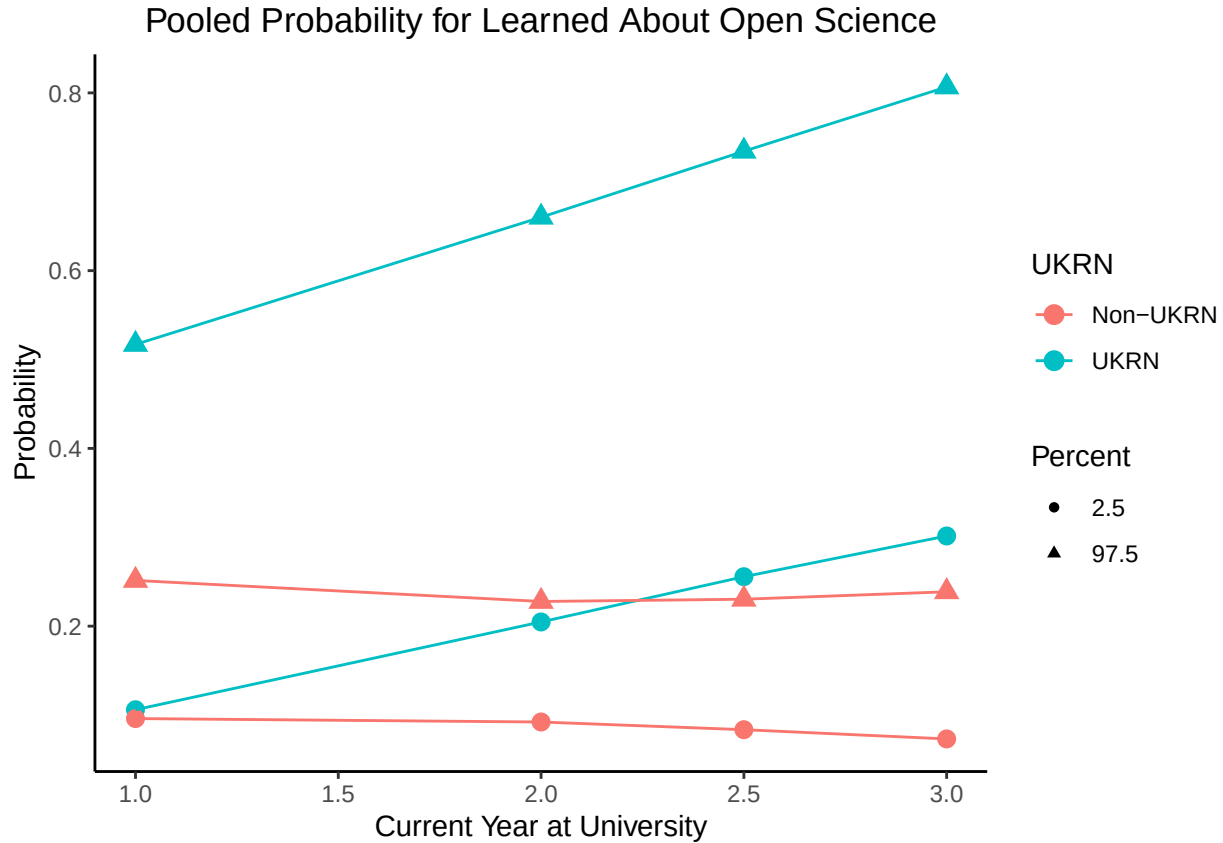
Placement Year N = 31

Year 4 N = 87

The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.140923  
 97.5% = 1.3398731

The interpretation is that students who attend a UKRN affiliated university are more likely to have learned about open science in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.



### Can explain open science

Bayesian Logistic regression for whether participants can explain what open science is (yes/no).

The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement of third year
- \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           -1.26    0.01 0.30   -1.86   -1.45   -1.25   -1.05   -0.69  1783
## sigma         0.37    0.00 0.15    0.09    0.26    0.36    0.47    0.71  1129
## alpha[1]     -1.36    0.01 0.42   -2.22   -1.62   -1.34   -1.08   -0.58  2564
## alpha[2]     -1.10    0.01 0.44   -1.95   -1.38   -1.10   -0.82   -0.19  3210
## alpha[3]     -1.32    0.01 0.42   -2.18   -1.58   -1.30   -1.04   -0.53  2596
## alpha[4]     -0.80    0.01 0.28   -1.35   -0.99   -0.80   -0.61   -0.24  1915
## alpha[5]     -1.14    0.01 0.43   -1.99   -1.42   -1.15   -0.87   -0.28  3335
## alpha[6]     -1.43    0.01 0.43   -2.35   -1.70   -1.40   -1.14   -0.64  2736
```

```

## alpha[7]      -1.41      0.01 0.40   -2.23   -1.67   -1.40   -1.14   -0.68  2589
## alpha[8]      -1.34      0.01 0.40   -2.16   -1.60   -1.32   -1.07   -0.60  2852
## alpha[9]      -1.54      0.01 0.37   -2.31   -1.78   -1.53   -1.28   -0.87  2191
## alpha[10]     -1.49      0.01 0.42   -2.36   -1.76   -1.47   -1.20   -0.74  2535
## alpha[11]     -1.36      0.01 0.44   -2.26   -1.63   -1.33   -1.06   -0.53  3478
## alpha[12]     -1.08      0.01 0.44   -1.93   -1.37   -1.09   -0.81   -0.17  3789
## alpha[13]     -1.28      0.01 0.41   -2.13   -1.55   -1.27   -1.01   -0.50  3353
## alpha[14]     -1.10      0.01 0.43   -1.95   -1.39   -1.11   -0.83   -0.22  3911
## alpha[15]     -1.36      0.01 0.40   -2.18   -1.62   -1.35   -1.09   -0.61  2808
## alpha[16]     -1.17      0.01 0.43   -2.03   -1.44   -1.17   -0.90   -0.33  3701
## alpha[17]     -1.51      0.01 0.46   -2.49   -1.80   -1.48   -1.20   -0.71  2589
## alpha[18]     -1.22      0.01 0.41   -2.06   -1.49   -1.22   -0.96   -0.41  3249
## alpha[19]     -0.84      0.01 0.44   -1.65   -1.14   -0.86   -0.56    0.10  2675
## beta1         0.07      0.01 0.60   -1.11   -0.34    0.06    0.46    1.26  3252
## beta2         0.13      0.00 0.15   -0.15    0.03    0.13    0.23    0.41  1967
## beta3         0.31      0.00 0.28   -0.22    0.13    0.31    0.50    0.86  3705
## lp__          -405.37    0.33 7.33 -418.07 -410.01 -405.95 -401.56 -387.40  492
##
## Rhat
## mu            1.00
## sigma         1.01
## alpha[1]      1.00
## alpha[2]      1.00
## alpha[3]      1.00
## alpha[4]      1.00
## alpha[5]      1.00
## alpha[6]      1.00
## alpha[7]      1.00
## alpha[8]      1.00
## alpha[9]      1.00
## alpha[10]     1.00
## alpha[11]     1.00
## alpha[12]     1.00
## alpha[13]     1.00
## alpha[14]     1.00
## alpha[15]     1.00
## alpha[16]     1.00
## alpha[17]     1.00
## alpha[18]     1.00
## alpha[19]     1.00
## beta1         1.00
## beta2         1.00
## beta3         1.00
## lp__          1.01
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 01:20:59 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

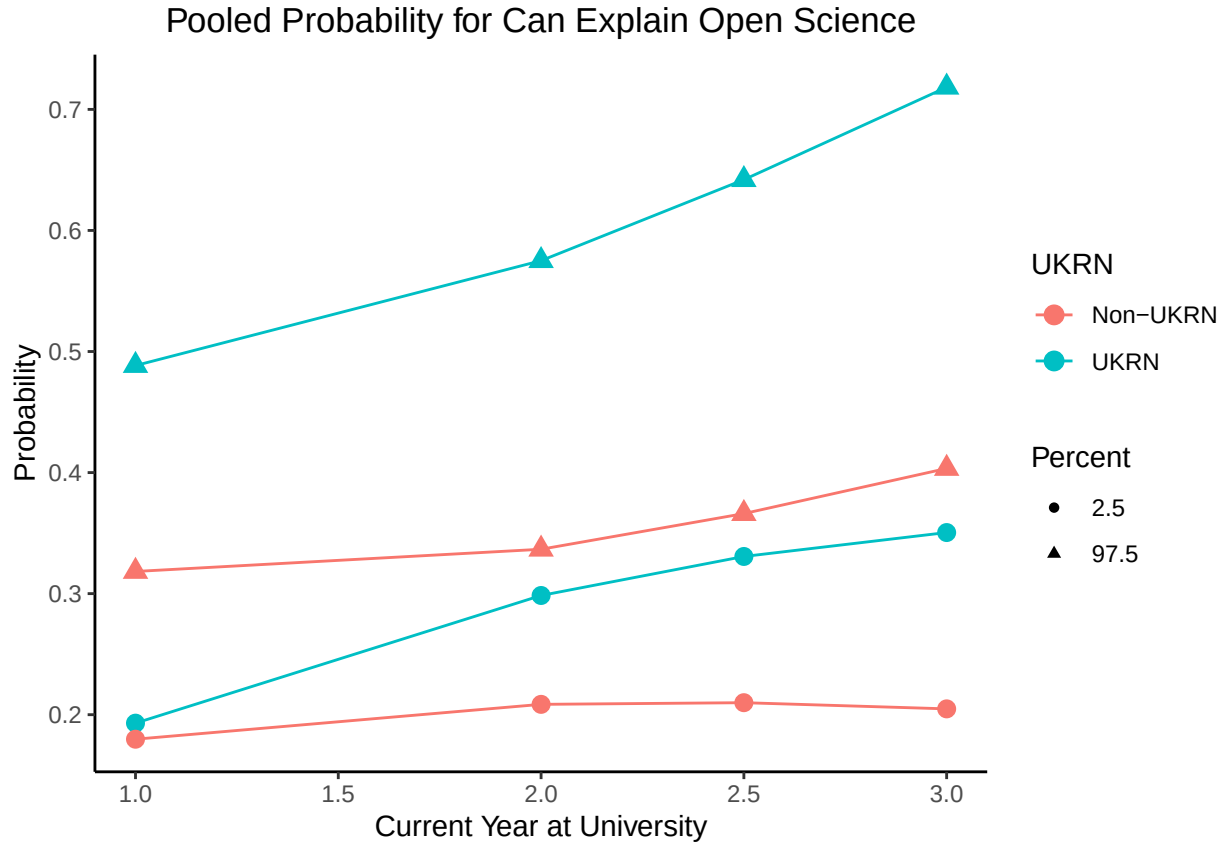
```

Number of participants in the analysis

Total N = 679  
 UKRN N = 220  
 Non-UKRN N = 459  
 Year 1 N = 308

Year 2 N = 253  
 Placement Year N = 31  
 Year 4 N = 87

The output shows that all beta credibility intervals do cross 0. No significant results.



## Applicability

Bayesian Logistic regression for whether participants think that the issues discussed in the questionnaire are relevant to them (yes/no). The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement or third year
- \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           2.13    0.01 0.45    1.28    1.82    2.11    2.42    3.06  2095
## sigma        0.68    0.00 0.24    0.30    0.50    0.64    0.81    1.24  2943
## alpha[1]     1.87    0.01 0.65    0.61    1.45    1.85    2.28    3.20  4182
## alpha[2]     2.66    0.01 0.85    1.22    2.08    2.59    3.16    4.53  3319
## alpha[3]     1.86    0.01 0.65    0.59    1.44    1.84    2.26    3.18  3959
## alpha[4]     1.12    0.01 0.33    0.47    0.89    1.12    1.34    1.78  3735
## alpha[5]     2.27    0.01 0.71    1.02    1.77    2.21    2.71    3.82  4467
## alpha[6]     2.20    0.01 0.64    1.04    1.76    2.17    2.61    3.56  4080
```

```

## alpha[7]      2.26      0.01 0.58      1.21      1.86      2.24      2.64      3.47      3547
## alpha[8]      2.29      0.01 0.60      1.18      1.87      2.26      2.67      3.52      3490
## alpha[9]      1.22      0.00 0.38      0.48      0.96      1.21      1.48      1.98      5898
## alpha[10]     2.00      0.01 0.56      0.98      1.62      1.97      2.36      3.16      4397
## alpha[11]     2.58      0.01 0.81      1.24      2.02      2.50      3.05      4.38      3650
## alpha[12]     2.52      0.01 0.80      1.16      1.97      2.44      3.00      4.31      4063
## alpha[13]     2.70      0.01 0.81      1.35      2.13      2.63      3.18      4.49      3153
## alpha[14]     2.22      0.01 0.70      0.96      1.74      2.17      2.64      3.74      4383
## alpha[15]     2.30      0.01 0.61      1.19      1.88      2.27      2.69      3.58      3542
## alpha[16]     2.00      0.01 0.64      0.82      1.57      1.97      2.41      3.34      5109
## alpha[17]     1.74      0.01 0.58      0.63      1.35      1.72      2.11      2.93      5270
## alpha[18]     2.34      0.01 0.69      1.13      1.86      2.30      2.77      3.81      4000
## alpha[19]     2.30      0.01 0.66      1.09      1.84      2.26      2.73      3.70      3496
## beta1         -1.00      0.01 0.85     -2.67     -1.56     -0.99     -0.44      0.68      5055
## beta2         -0.08      0.00 0.19     -0.46     -0.21     -0.08      0.05      0.30      2569
## beta3          0.61      0.00 0.38     -0.12      0.36      0.61      0.87      1.36      5861
## lp__          -299.55      0.11 5.49  -310.79  -303.10  -299.39  -295.84  -289.23      2601
##
## Rhat
## mu            1
## sigma         1
## alpha[1]      1
## alpha[2]      1
## alpha[3]      1
## alpha[4]      1
## alpha[5]      1
## alpha[6]      1
## alpha[7]      1
## alpha[8]      1
## alpha[9]      1
## alpha[10]     1
## alpha[11]     1
## alpha[12]     1
## alpha[13]     1
## alpha[14]     1
## alpha[15]     1
## alpha[16]     1
## alpha[17]     1
## alpha[18]     1
## alpha[19]     1
## beta1         1
## beta2         1
## beta3         1
## lp__          1
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 01:23:48 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 679

UKRN N = 220

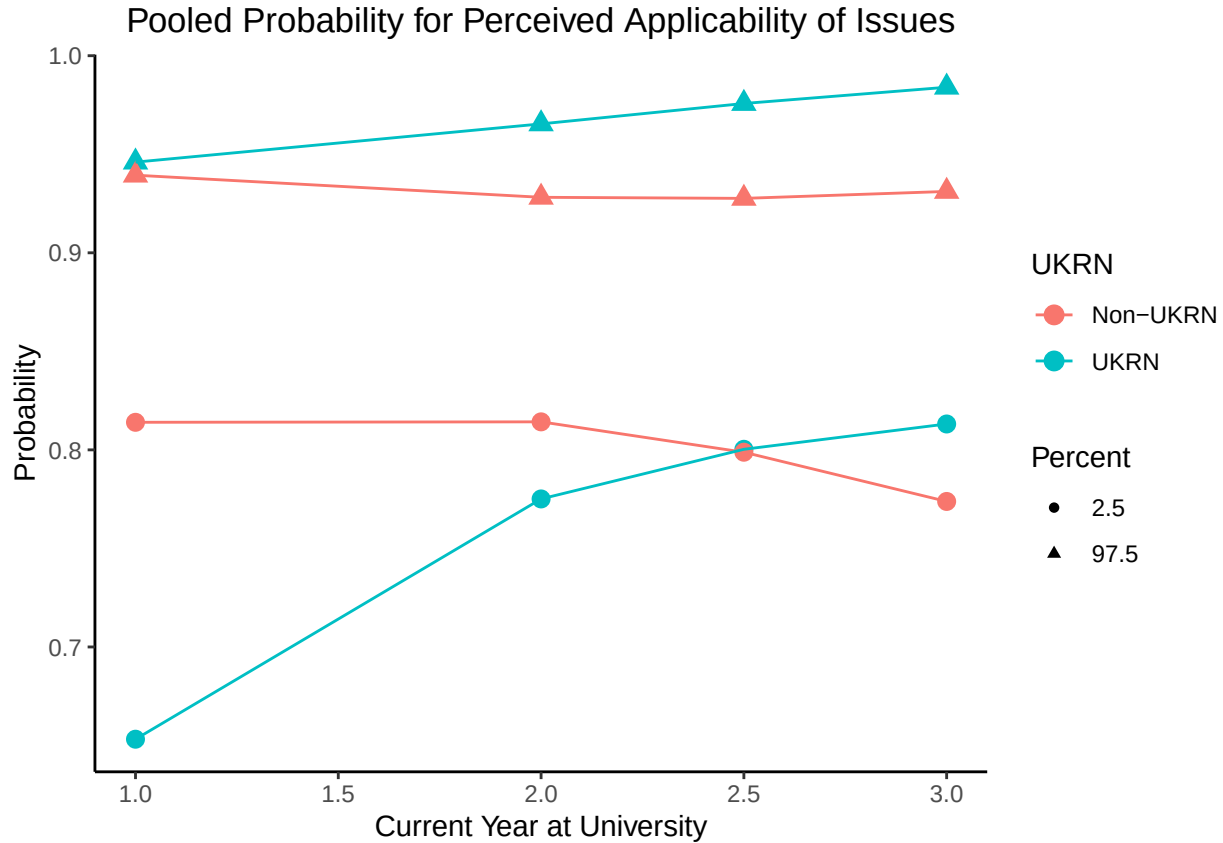
Non-UKRN N = 459

Year 1 N = 308



Year 2 N = 253  
 Placement Year N = 31  
 Year 4 N = 87

The output shows that all beta credibility intervals do cross 0. No significant results.



## Data Set 2 - Truncated Poisson Regression

### Awarweness Score Total

Bayesian regression for how much awareness of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement or third year
- \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean  sd   2.5%   25%   50%   75%   97.5% n_eff
## mu         1.40     0.00 0.07   1.27   1.36   1.41   1.45   1.54  2025
## sigma      0.09     0.00 0.04   0.02   0.07   0.09   0.11   0.17   302
## alpha[1]    1.42     0.00 0.10   1.22   1.35   1.42   1.48   1.61  3219
## alpha[2]    1.42     0.00 0.11   1.21   1.35   1.42   1.48   1.63  3849
## alpha[3]    1.38     0.00 0.10   1.18   1.32   1.39   1.45   1.57  3043
## alpha[4]    1.53     0.00 0.07   1.40   1.48   1.53   1.57   1.66   787
```

```

## alpha[5]      1.46      0.00 0.10      1.26      1.39      1.45      1.52      1.67 3834
## alpha[6]      1.35      0.00 0.10      1.15      1.29      1.36      1.42      1.54 3277
## alpha[7]      1.35      0.00 0.09      1.17      1.29      1.36      1.42      1.53 2251
## alpha[8]      1.41      0.00 0.09      1.23      1.35      1.41      1.47      1.59 3381
## alpha[9]      1.29      0.00 0.09      1.12      1.24      1.30      1.36      1.45  955
## alpha[10]     1.38      0.00 0.09      1.19      1.32      1.38      1.43      1.55 3352
## alpha[11]     1.41      0.00 0.10      1.21      1.34      1.41      1.47      1.60 4240
## alpha[12]     1.48      0.00 0.11      1.29      1.41      1.47      1.55      1.72 3630
## alpha[13]     1.39      0.00 0.10      1.20      1.33      1.39      1.45      1.58 3808
## alpha[14]     1.44      0.00 0.10      1.24      1.37      1.43      1.50      1.64 3833
## alpha[15]     1.39      0.00 0.09      1.21      1.33      1.39      1.45      1.57 3163
## alpha[16]     1.40      0.00 0.10      1.20      1.34      1.40      1.47      1.60 4070
## alpha[17]     1.40      0.00 0.10      1.20      1.34      1.40      1.46      1.59 3830
## alpha[18]     1.39      0.00 0.10      1.19      1.33      1.40      1.46      1.58 3985
## alpha[19]     1.40      0.00 0.10      1.20      1.34      1.40      1.46      1.59 3519
## beta1         0.06      0.00 0.15     -0.23     -0.04      0.06      0.16      0.34 2251
## beta2         0.05      0.00 0.03     -0.02      0.03      0.05      0.07      0.12 2178
## beta3         0.08      0.00 0.07     -0.05      0.03      0.08      0.13      0.22 1145
## lp__          1681.70    0.57 6.65 1669.72 1677.53 1681.32 1685.21 1699.16   134
##
## Rhat
## mu            1.00
## sigma         1.02
## alpha[1]      1.00
## alpha[2]      1.00
## alpha[3]      1.00
## alpha[4]      1.01
## alpha[5]      1.00
## alpha[6]      1.00
## alpha[7]      1.00
## alpha[8]      1.00
## alpha[9]      1.01
## alpha[10]     1.00
## alpha[11]     1.00
## alpha[12]     1.00
## alpha[13]     1.00
## alpha[14]     1.00
## alpha[15]     1.00
## alpha[16]     1.00
## alpha[17]     1.00
## alpha[18]     1.00
## alpha[19]     1.00
## beta1         1.00
## beta2         1.00
## beta3         1.00
## lp__          1.04
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 01:38:03 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 696

UKRN N = 222

Non-UKRN N = 474

Year 1 N = 316

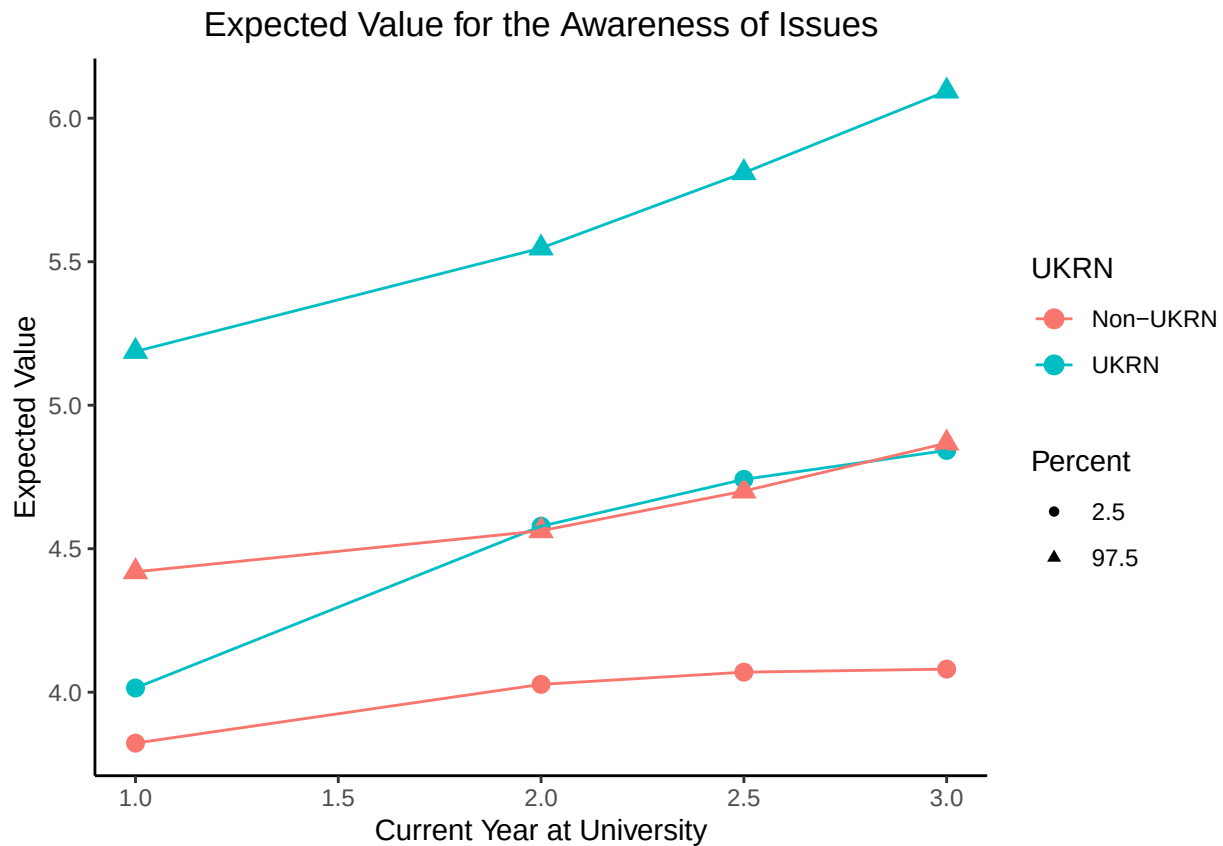
Year 2 N = 258

Placement Year N = 31

Year 4 N = 91

| Var1 | Freq |
|------|------|
| 0    | 18   |
| 1    | 39   |
| 2    | 57   |
| 3    | 110  |
| 4    | 135  |
| 5    | 114  |
| 6    | 88   |
| 7    | 59   |
| 8    | 76   |

The output shows that all beta credibility intervals do cross 0. No significant results.



### Experience Score Total

Bayesian regression for how much experience of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

\* Whether the participant was at a university that was affiliated with the UKRN

\* Whether the participant was in their first, second, placement or third year

\* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean    sd    2.5%    25%    50%    75%    97.5% n_eff
## mu            0.68     0.00  0.08    0.53    0.62    0.68    0.74    0.84   475
## sigma         0.08     0.00  0.05    0.01    0.04    0.06    0.10    0.21   377
## alpha[1]      0.68     0.00  0.10    0.48    0.61    0.68    0.75    0.89   986
## alpha[2]      0.67     0.00  0.11    0.45    0.61    0.67    0.74    0.89  1435
## alpha[3]      0.68     0.01  0.11    0.49    0.62    0.68    0.75    0.90   369
## alpha[4]      0.68     0.00  0.08    0.54    0.63    0.68    0.73    0.84   340
## alpha[5]      0.75     0.01  0.13    0.56    0.66    0.74    0.82    1.06   501
## alpha[6]      0.73     0.01  0.11    0.54    0.65    0.72    0.79    0.97   390
## alpha[7]      0.67     0.00  0.10    0.48    0.60    0.67    0.74    0.87   790
## alpha[8]      0.72     0.01  0.10    0.55    0.65    0.72    0.79    0.95   358
## alpha[9]      0.69     0.00  0.08    0.53    0.63    0.69    0.75    0.87   732
## alpha[10]     0.68     0.00  0.10    0.50    0.61    0.68    0.75    0.88   595
## alpha[11]     0.69     0.00  0.10    0.49    0.62    0.68    0.76    0.91   681
## alpha[12]     0.61     0.00  0.13    0.29    0.55    0.62    0.69    0.82  1463
## alpha[13]     0.71     0.01  0.11    0.52    0.63    0.70    0.77    0.94   371
## alpha[14]     0.61     0.00  0.13    0.31    0.55    0.62    0.70    0.82  1603
## alpha[15]     0.67     0.00  0.10    0.47    0.61    0.67    0.73    0.86  1381
## alpha[16]     0.65     0.00  0.11    0.40    0.59    0.65    0.72    0.86  1815
## alpha[17]     0.69     0.00  0.10    0.49    0.62    0.68    0.75    0.89  1141
## alpha[18]     0.65     0.00  0.11    0.41    0.59    0.65    0.72    0.85  1588
## alpha[19]     0.69     0.00  0.10    0.49    0.62    0.68    0.75    0.90   807
## beta1         -0.30    0.01  0.18   -0.66   -0.42   -0.29   -0.17    0.05  1012
## beta2          0.09    0.00  0.04    0.01    0.07    0.09    0.12    0.17   127
## beta3          0.04    0.00  0.09   -0.13   -0.02    0.04    0.10    0.21   781
## lp__          -281.70    0.98 11.21 -300.59 -290.43 -282.97 -273.67 -260.13  131
##
##               Rhat
## mu            1.01
## sigma         1.02
## alpha[1]      1.01
## alpha[2]      1.01
## alpha[3]      1.01
## alpha[4]      1.01
## alpha[5]      1.02
## alpha[6]      1.02
## alpha[7]      1.01
## alpha[8]      1.02
## alpha[9]      1.01
## alpha[10]     1.01
## alpha[11]     1.01
## alpha[12]     1.00
## alpha[13]     1.02
## alpha[14]     1.00
## alpha[15]     1.00
## alpha[16]     1.00
## alpha[17]     1.01
## alpha[18]     1.00
```

```
## alpha[19] 1.01
## beta1     1.00
## beta2     1.03
## beta3     1.01
## lp__      1.04
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 10:42:58 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

#### Number of participants in the analysis

Total N = 696  
 UKRN N = 222  
 Non-UKRN N = 474  
 Year 1 N = 316  
 Year 2 N = 258  
 Placement Year N = 31  
 Year 4 N = 91

| Var1 | Freq |
|------|------|
| 0    | 156  |
| 1    | 110  |
| 2    | 149  |
| 3    | 130  |
| 4    | 73   |
| 5    | 52   |
| 6    | 15   |
| 7    | 10   |
| 8    | 1    |

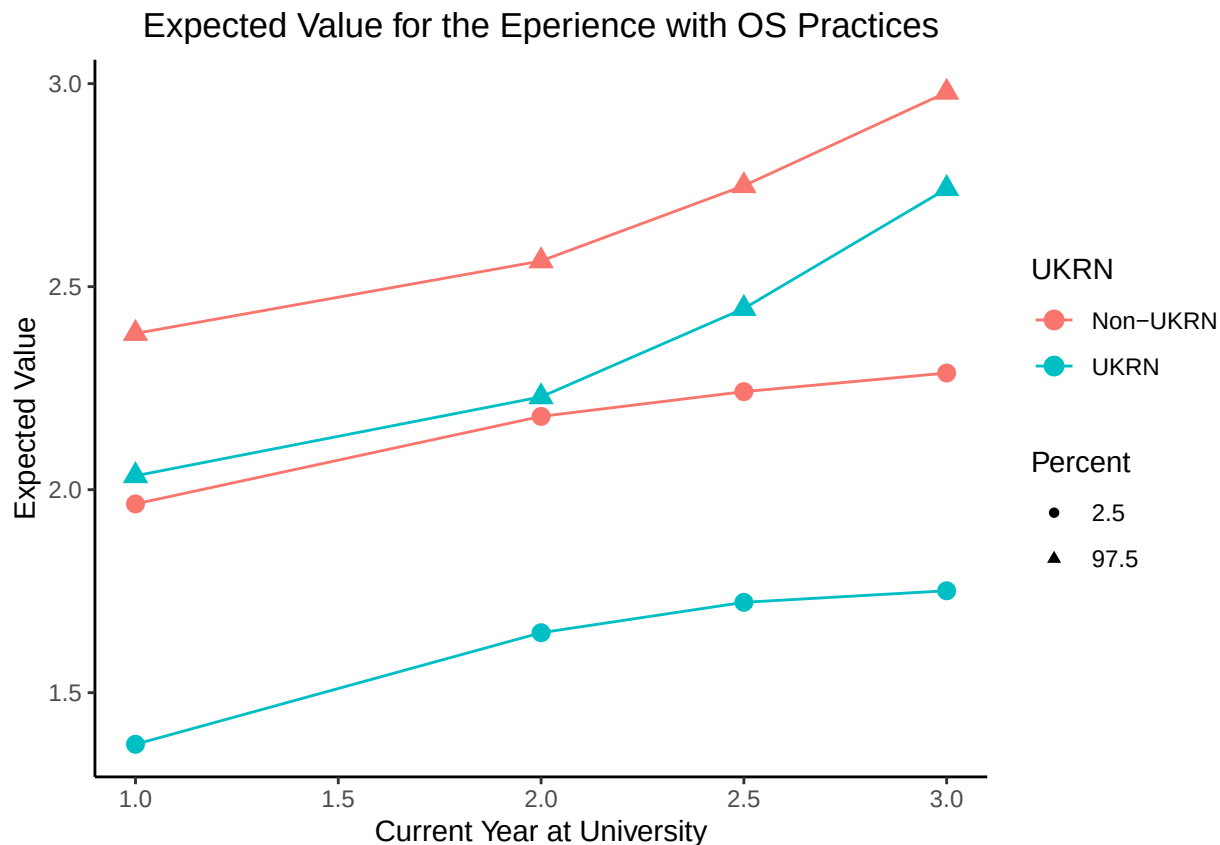
The output shows that all beta credibility intervals cross 0.

The output shows that beta credibility intervals do not cross 0 only for the effect of year of study.

2.5% = 0.0133349

97.5% = 0.1681478

The interpretation is that students are more likely to have experience with open science practices in the later years of education regardless of whether they are in a UKRN or non-UKRN university.



## Data Set 2 - Linear Regression

### Conceptual Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research.

The explainer variables are:

\* Whether the participant was at a university that was affiliated with the UKRN

\* Whether the participant was in their first, second, placement or third year

\* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
```

```
## 4 chains, each with iter=10000; warmup=5000; thin=1;
```

```
## post-warmup draws per chain=5000, total post-warmup draws=20000.
```

```
##
```

|             | mean | se_mean | sd   | 2.5% | 25%  | 50%  | 75%  | 97.5% | n_eff | Rhat |
|-------------|------|---------|------|------|------|------|------|-------|-------|------|
| ## mu       | 0.48 | 0.00    | 0.05 | 0.39 | 0.45 | 0.48 | 0.52 | 0.58  | 1852  | 1    |
| ## sigma    | 0.37 | 0.00    | 0.01 | 0.35 | 0.36 | 0.37 | 0.37 | 0.39  | 12308 | 1    |
| ## sigma_a  | 0.07 | 0.00    | 0.03 | 0.02 | 0.05 | 0.06 | 0.08 | 0.13  | 2078  | 1    |
| ## alpha[1] | 0.43 | 0.00    | 0.07 | 0.28 | 0.38 | 0.43 | 0.47 | 0.57  | 3275  | 1    |
| ## alpha[2] | 0.50 | 0.00    | 0.08 | 0.35 | 0.45 | 0.50 | 0.55 | 0.67  | 3063  | 1    |
| ## alpha[3] | 0.52 | 0.00    | 0.07 | 0.39 | 0.47 | 0.52 | 0.57 | 0.68  | 2523  | 1    |
| ## alpha[4] | 0.43 | 0.00    | 0.05 | 0.34 | 0.40 | 0.43 | 0.46 | 0.52  | 2848  | 1    |
| ## alpha[5] | 0.56 | 0.00    | 0.09 | 0.40 | 0.49 | 0.55 | 0.61 | 0.75  | 2341  | 1    |
| ## alpha[6] | 0.52 | 0.00    | 0.07 | 0.38 | 0.47 | 0.51 | 0.56 | 0.67  | 2870  | 1    |
| ## alpha[7] | 0.49 | 0.00    | 0.06 | 0.36 | 0.44 | 0.49 | 0.53 | 0.62  | 2776  | 1    |
| ## alpha[8] | 0.48 | 0.00    | 0.06 | 0.36 | 0.44 | 0.48 | 0.53 | 0.61  | 2953  | 1    |
| ## alpha[9] | 0.45 | 0.00    | 0.05 | 0.34 | 0.41 | 0.45 | 0.48 | 0.55  | 4258  | 1    |

```

## alpha[10]    0.48    0.00 0.06    0.35    0.44    0.48    0.52    0.61  3393    1
## alpha[11]    0.46    0.00 0.07    0.31    0.41    0.46    0.51    0.60  4324    1
## alpha[12]    0.55    0.00 0.09    0.40    0.48    0.54    0.60    0.74  2615    1
## alpha[13]    0.44    0.00 0.07    0.30    0.40    0.44    0.49    0.57  4375    1
## alpha[14]    0.48    0.00 0.07    0.34    0.43    0.48    0.53    0.63  3722    1
## alpha[15]    0.47    0.00 0.06    0.35    0.43    0.47    0.52    0.60  3133    1
## alpha[16]    0.46    0.00 0.07    0.31    0.41    0.46    0.50    0.60  4476    1
## alpha[17]    0.43    0.00 0.07    0.28    0.38    0.43    0.48    0.56  4481    1
## alpha[18]    0.50    0.00 0.07    0.37    0.45    0.50    0.55    0.66  3245    1
## alpha[19]    0.52    0.00 0.07    0.38    0.47    0.51    0.56    0.67  2659    1
## beta1        -0.09    0.00 0.10   -0.29   -0.15   -0.09   -0.02    0.11  3584    1
## beta2         0.05    0.00 0.02    0.00    0.03    0.05    0.07    0.10  2099    1
## beta3         0.07    0.00 0.05   -0.02    0.04    0.07    0.10    0.16  3985    1
## lp__         390.28    0.19 6.71 378.14 385.81 389.91 394.15 405.50 1231    1
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 11:01:31 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

#### Number of participants in the analysis

Total N = 690

UKRN N = 221

Non-UKRN N = 469

Year 1 N = 311

Year 2 N = 257

Placement Year N = 31

Year 4 N = 91

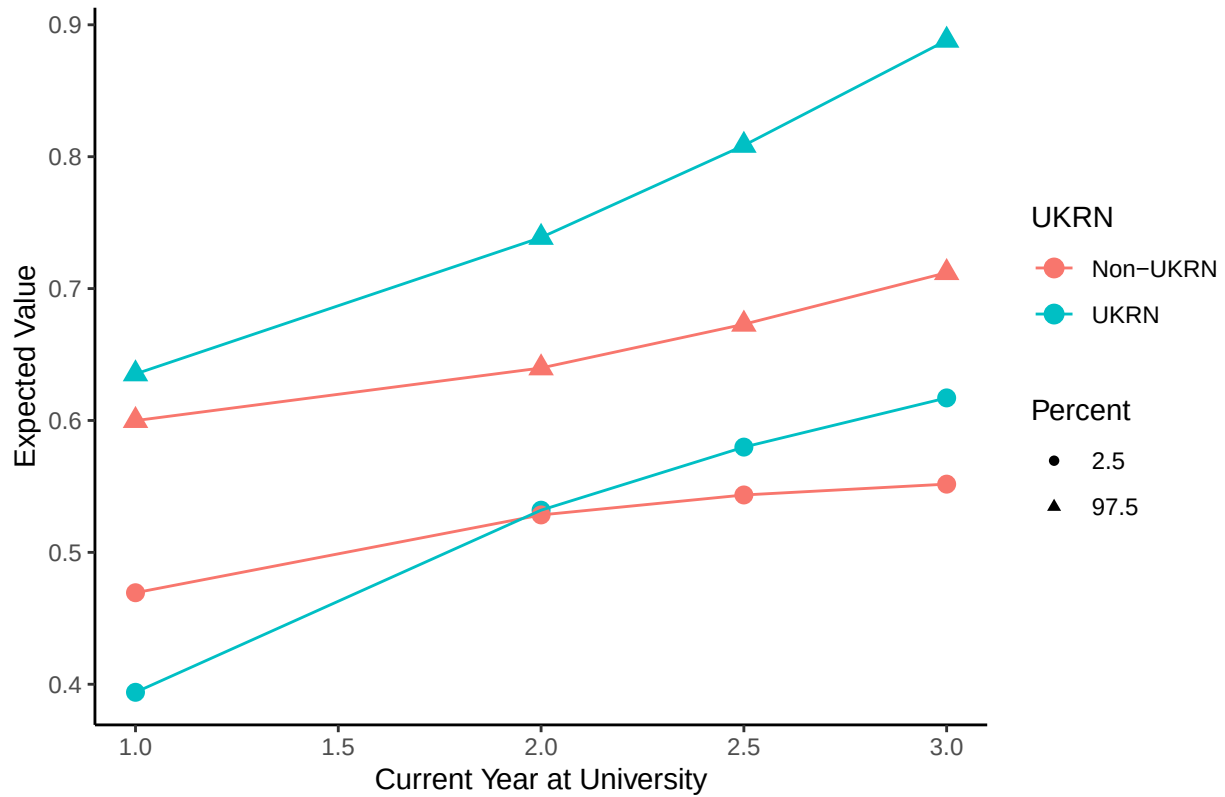
The output shows that beta credibility intervals do not cross 0 only for the effect of year of study.

2.5% = 0.0029804

97.5% = 0.0972215

The interpretation is that students are more likely to have positive perceptions of open science practices in the later years of education regardless of whether they are in a UKRN or non-UKRN university.

## Expected Value for positive Conceptual Perceptions of OS Practices



## Situational Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research.

The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement or third year
- \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff Rhat
## mu           0.10    0.00  0.06  -0.01  0.06  0.10  0.15  0.23  1343 1.00
## sigma        0.50    0.00  0.01   0.47  0.49  0.50  0.51  0.53  3473 1.00
## sigma_a      0.06    0.00  0.04   0.01  0.03  0.06  0.08  0.15   760 1.00
## alpha[1]     0.07    0.00  0.08  -0.11  0.02  0.07  0.12  0.22  2184 1.00
## alpha[2]     0.11    0.00  0.09  -0.05  0.06  0.11  0.17  0.29  1940 1.00
## alpha[3]     0.13    0.00  0.08  -0.02  0.08  0.13  0.18  0.30  1739 1.00
## alpha[4]     0.08    0.00  0.06  -0.04  0.04  0.08  0.12  0.19  1841 1.00
## alpha[5]     0.16    0.00  0.10   0.00  0.09  0.15  0.21  0.37  1347 1.00
## alpha[6]     0.13    0.00  0.08  -0.03  0.07  0.12  0.18  0.31  1797 1.00
## alpha[7]     0.10    0.00  0.08  -0.05  0.05  0.10  0.15  0.25  1977 1.00
## alpha[8]     0.12    0.00  0.08  -0.03  0.06  0.11  0.17  0.28  1808 1.00
## alpha[9]     0.12    0.00  0.07  -0.01  0.08  0.12  0.17  0.26  1390 1.00
## alpha[10]    0.10    0.00  0.08  -0.05  0.05  0.10  0.15  0.25  2124 1.00
```



```
## alpha[11]  0.10    0.00  0.08 -0.06   0.05   0.10   0.16   0.27  2262 1.00
## alpha[12]  0.16    0.00  0.10   0.00   0.09   0.15   0.21   0.38  1310 1.00
## alpha[13]  0.05    0.00  0.09 -0.15   0.00   0.06   0.11   0.21  1975 1.00
## alpha[14]  0.07    0.00  0.09 -0.12   0.02   0.08   0.13   0.23  2617 1.00
## alpha[15]  0.08    0.00  0.08 -0.08   0.03   0.08   0.13   0.23  2281 1.00
## alpha[16]  0.11    0.00  0.08 -0.05   0.06   0.11   0.16   0.28  2321 1.00
## alpha[17]  0.08    0.00  0.08 -0.09   0.03   0.09   0.14   0.24  2257 1.00
## alpha[18]  0.09    0.00  0.08 -0.08   0.04   0.09   0.15   0.25  2552 1.00
## alpha[19]  0.11    0.00  0.08 -0.04   0.06   0.11   0.16   0.28  2145 1.00
## beta1      0.06    0.00  0.13 -0.19 -0.02   0.06   0.15   0.31  2878 1.00
## beta2      0.14    0.00  0.03  0.08   0.12   0.14   0.16   0.20  1285 1.00
## beta3      0.01    0.00  0.06 -0.11 -0.03   0.01   0.05   0.13  2762 1.00
## lp__       178.87    0.54 10.20 162.21 171.67 177.42 184.84 202.15   361 1.01
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 11:22:24 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

#### Number of participants in the analysis

**Total N** = 682

**UKRN N** = 221

**Non-UKRN N** = 461

**Year 1 N** = 308

**Year 2 N** = 255

**Placement Year N** = 31

**Year 4 N** = 88

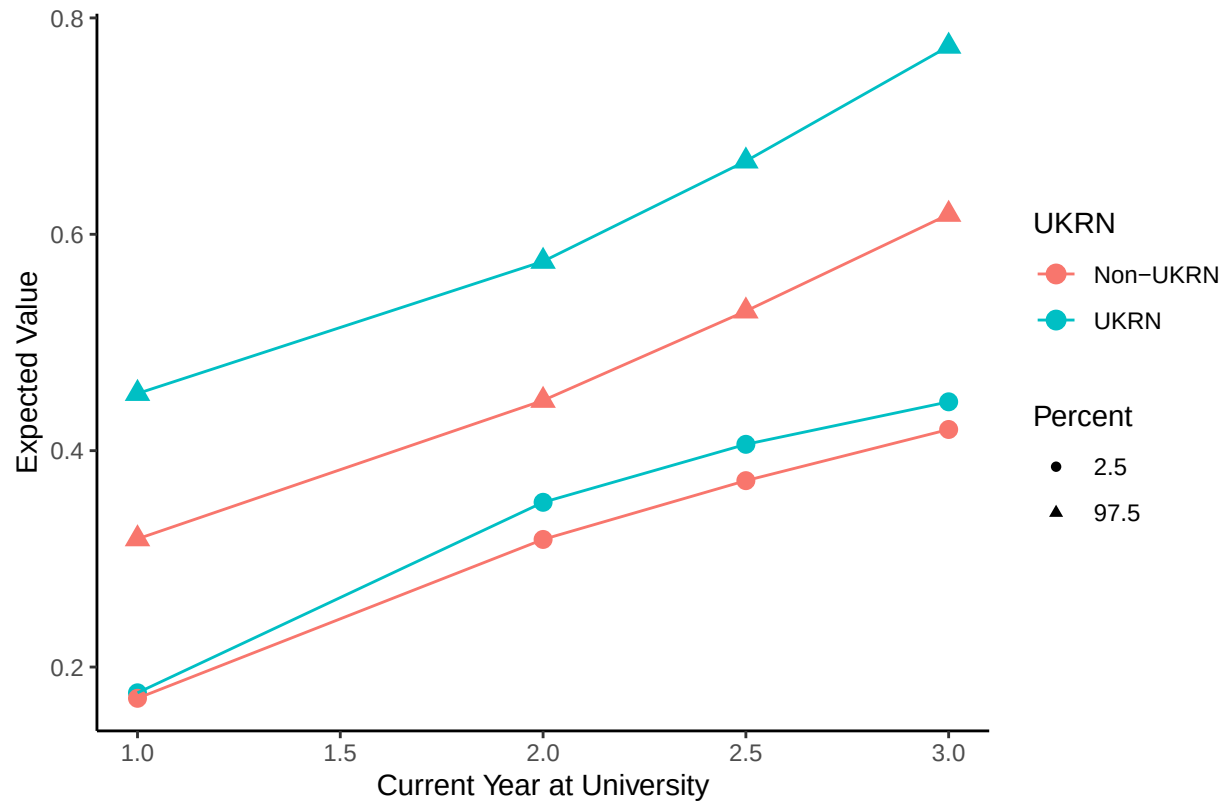
The output shows that beta credibility intervals do not cross 0 only for the effect of year of study.

2.5% = 0.0781992

97.5% = 0.1997335

The interpretation is that students are more likely to have positive perceptions of open science practices in the later years of education regardless of whether they are in a UKRN or non-UKRN university.

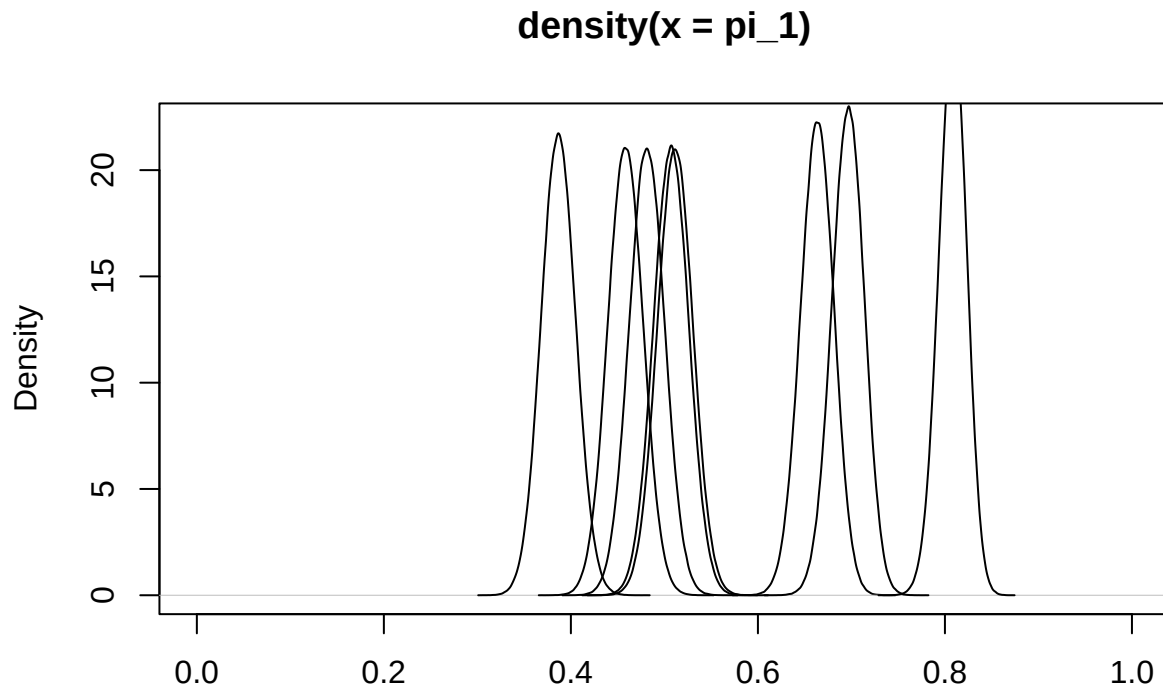
Expected Value for positive Conceptual Perceptions of OS Practices



## Dataset 2 - Beta Bernoulli Model

For the most common category in the Awareness variable

The categories are: \* Replication \* Open Research Materials \* Open Educational Materials \* Open Access Journals \* Open Data \* Power Analysis \* Preregistration \* Preprint



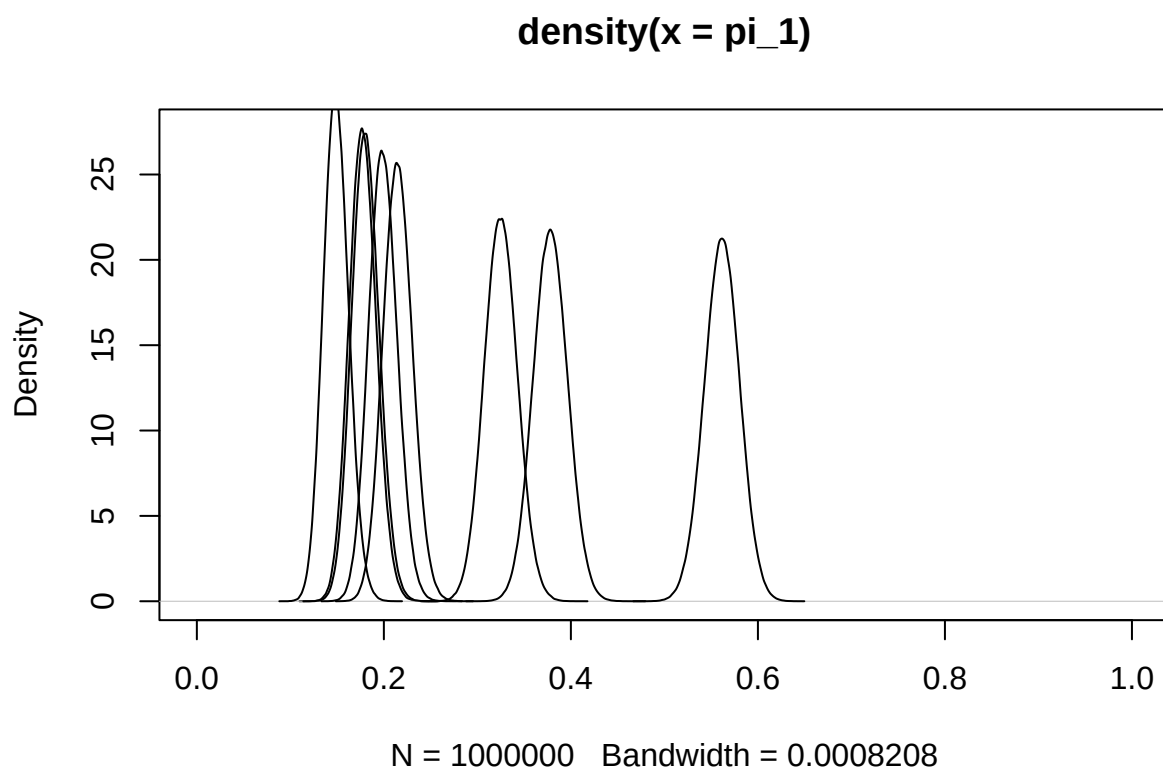
N = 1000000 Bandwidth = 0.001015

| ##                            | Probability |
|-------------------------------|-------------|
| ## Replication                | 0.000000    |
| ## Open Research Materials    | 0.000000    |
| ## Open Educational Materials | 0.000001    |
| ## Open Access Journals       | 0.000000    |
| ## Open Data                  | 0.000000    |
| ## Power Analysis             | 0.000000    |
| ## Preregistration            | 0.000000    |
| ## Preprint                   | 0.999999    |

probs\_Awareness

**For the most common category in the Experience variable**

The categories are: \* Replication \* Open Research Materials \* Open Educational Materials \* Open Access Journals \* Open Data \* Power Analysis \* Preregistration \* Preprint



| ##                             | Probability |
|--------------------------------|-------------|
| ## Replication                 | 0           |
| ## Open Research Materials     | 0           |
| ## Open Educational\nMaterials | 1           |
| ## Open Access Journals        | 0           |
| ## Open Data                   | 0           |
| ## Power Analysis              | 0           |
| ## Preregistration             | 0           |
| ## Preprint                    | 0           |

probs\_Experience